P1 Field inside solenoid with n turns per unit length and current I: $B = \mu_0 n I$

(i) $\vec{B}$ inside inner solenoid is along the axis, ($+\hat{z}$ is to the right)

$$\vec{B} = \mu_0 (n_1 - n_2) I \hat{z}$$

(ii) between solenoids $\vec{B} = -\mu_0 n_2 I \hat{z}$

(iii) outside: $\vec{B} = 0$

~~We can fix constant d by requiring $\vec{A} = A_z \hat{z}$ be continuous at $s = R$:~~

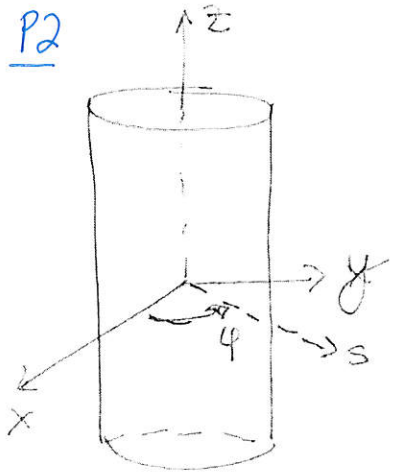
~~from part (a): $A_z = \frac{\mu_0 I}{2\pi} \ln\left(\frac{2L}{s}\right)\Big|_{s=R} = \frac{\mu_0 I}{2\pi} \ln\left(\frac{2L}{R}\right)$.~~

~~from (b): $A_z(s=R) = \frac{\mu_0 I}{4\pi}\left(\frac{d^2}{R^2} - 1\right)$~~

$$\Rightarrow \cancel{\frac{\mu_0 I}{4\pi}\left(\frac{d^2}{R^2} - 1\right) = \frac{\mu_0 I}{2\pi} \ln\left(\frac{2L}{R}\right)}$$

$$\cancel{\frac{d^2}{R^2} = 1 + 2\ln\left(\frac{2L}{R}\right)}.$$

~~P6.8~~ P2



$$\vec{M} = k s^2 \hat{\varphi} = M_\varphi \hat{\varphi}.$$

Volume magnetization current

$$\vec{J}_b = \vec{\nabla} \times \vec{M} = -\frac{\partial M_\varphi}{\partial z} \hat{s} + \frac{1}{s}\frac{\partial}{\partial s}(s M_\varphi)\hat{z}$$

$$\Rightarrow \vec{J}_b = \frac{1}{s}\frac{\partial}{\partial s}(k s^3)\hat{z} = \frac{3ks^2}{s} = 3ks\,\hat{z}$$

Surface magnetization current $\vec{K} = \vec{M} \times \hat{n} = -k s^2 \hat{z}\Big|_{s=R}$

$\hat{n} = \hat{s}$, $\hat{\varphi} \times \hat{s} = -\hat{z}$

$$\vec{K} = -kR^2 \hat{z}.$$

$\Rightarrow$ surface current flows <u>opposite</u> to <u>bulk</u>.

So the problem is now equivalent to finding $\vec{B}$ due to currents $\vec{J}_b$ and $\vec{K}_b$.

Inside the cylinder $(r<R)$ $\quad \vec{B} = B\hat{\varphi}$

$$B \cdot 2\pi r = \oint \vec{B}\cdot d\vec{l} = \mu_0 I_{enc} = \mu_0 \int_0^r (3kr)\, 2\pi r\, dr =$$

$$= \mu_0\, 6\pi k \int_0^r r^2 dr = \mu_0\, 2\pi k\, r^3.$$

$$\Rightarrow B = \frac{2\pi k \mu_0 r^3}{2\pi r} = \mu_0 \underbrace{k r^2}_{M.} \Rightarrow \vec{B} = \mu_0 \vec{M}.$$

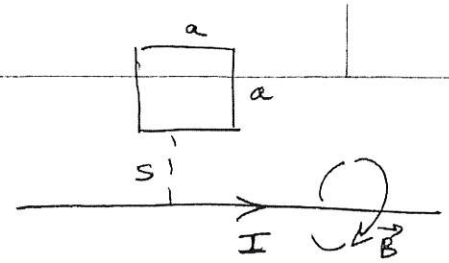
<u>$r>R$</u> Outside the cylinder we need total current enclosed:

$$I_{enc} = \int_0^R (3kr)\, 2\pi r\, dr + \underbrace{(-kR^2)(2\pi R)}_{\text{surface current}} =$$

$$= 6\pi k \int_0^R r^2 dr - 2\pi k R^3 = 0$$

$\Rightarrow B = 0$ outside.

~~P8~~ P3



$$dA = a\,d\rho$$

(a) $\vec{B} = \frac{\mu_0 I}{2\pi\rho}\hat{\varphi} \Rightarrow \phi = \int \frac{\mu_0 I}{2\pi\rho}\,dA = \frac{\mu_0 I a}{2\pi}\int_s^{s+a}\frac{d\rho}{\rho} =$

$$= \frac{\mu_0 I a}{2\pi}\ln\left(\frac{s+a}{s}\right)$$

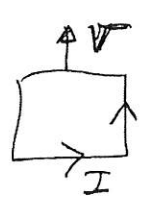
(b) The loop is pulled away $\Rightarrow$ $s(t) = s_0 + vt$

$$\Rightarrow \phi(t) = \frac{\mu_0 I a}{2\pi}\ln\left(1+\frac{a}{s_0+vt}\right) \xrightarrow[t\to\infty]{} 0$$

$$\mathcal{E} = -\frac{d\phi}{dt} = -\frac{\mu_0 I a}{2\pi}\cdot\frac{1}{1+\frac{a}{s_0+vt}}\cdot\frac{-av}{(s_0+vt)^2} =$$

$$= \frac{\mu_0 I a^2 v}{2\pi}\,\frac{1}{(s_0+vt)(s_0+vt+a)}$$

Flux (which points out of the page) is diminishing with time t $\Rightarrow$ generated current in the loop flows counterclockwise in order to support it.



(c) If $\vec{v}$ is along the wire that s does not change

$$\Rightarrow \frac{d\phi}{dt} = 0 \Rightarrow \mathcal{E} = 0.$$

~~P10~~

top view B

$\phi = B\cdot a^2\cdot\cos\theta = Ba^2\cos(\omega t+\theta_0) \Rightarrow$

$\theta = \omega t + \theta_0$ $\quad \mathcal{E}(t) = -\frac{d\phi}{dt} = Ba^2\omega\sin(\omega t+\theta_0)$

P4 RLC circuit

$$I(t) = \frac{V_0}{|Z|} \cos(\omega t - \varphi)$$

$$|Z| = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2} = \sqrt{R^2 + \frac{1}{\omega^2 C^2}\left(\omega^2 LC - 1\right)^2} =$$

$$= \sqrt{R^2 + \frac{1}{\omega^2 C^2}\left(\frac{\omega^2}{\omega_0^2} - 1\right)^2}$$

At $\omega_1 = \omega_0 = \frac{1}{\sqrt{LC}}$

$|Z(\omega_1)| = R$ and $I_0(\omega_1) = \frac{V_0}{|Z|} = \frac{V_0}{R}$

At $\omega_2 = \sqrt{2}\,\omega_0 = \frac{\sqrt{2}}{\sqrt{LC}}$ $\quad |Z(\omega_2)| = \sqrt{R^2 + \frac{LC}{2C^2}(2-1)^2} =$

$$= \sqrt{R^2 + \frac{1}{2}\frac{L}{C}}$$

$$\Rightarrow I_0(\omega_2) = \frac{V_0}{\sqrt{R^2 + \frac{L}{2C}}}$$

$$\frac{I_0(\omega_1)}{I_0(\omega_2)} = \frac{\sqrt{R^2 + \frac{L}{2C}}}{R} = \sqrt{1 + \frac{L}{2CR^2}} = 2$$

$$\Rightarrow 1 + \frac{L}{2CR^2} = 4 \Rightarrow \frac{L}{CR^2} = 6.$$

$$R = \sqrt{\frac{L}{6C}}.$$